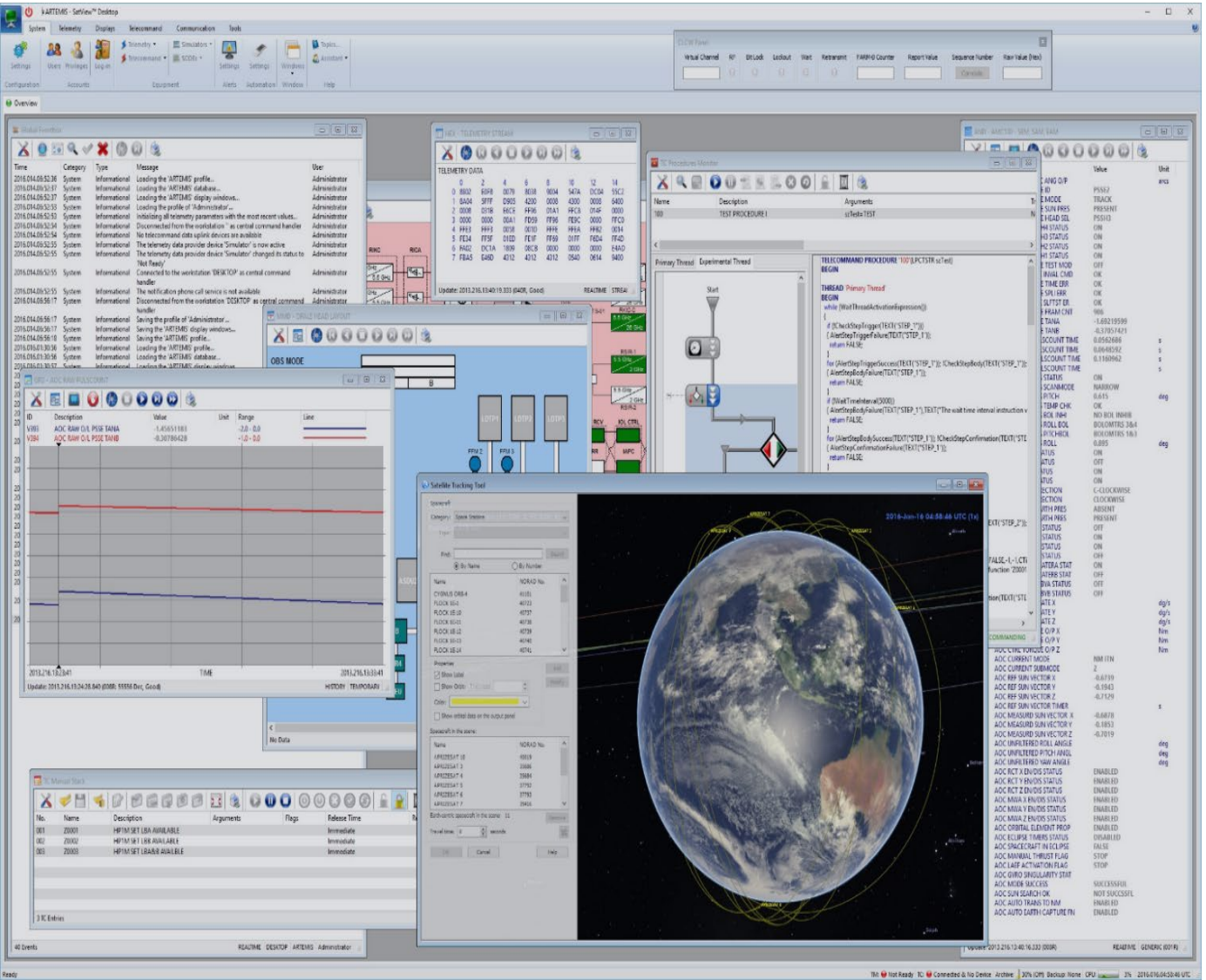
Satview Desktop



SatView (specifically the hARTEMIS SatView Desktop) is an advanced, all-in-one Satellite Monitoring and Control System used by aerospace engineers—including those at the European Space Agency (ESA)—for real-time mission operations. Designed as a comprehensive "mission control in a box," it integrates everything an operator needs into a single environment: 3D orbital tracking, live telemetry data streams, system health graphs, and telecommand scripting. The user experience is highly technical and data-dense, requiring operators to manage multiple floating windows simultaneously. While the interface might look overwhelming to an outsider, this customizable layout is intentionally built for operational efficiency; it ensures that engineers can instantly correlate raw spacecraft data with visual location tracking and execute critical commands without ever having to switch to a different software program.

Key Capabilities:

I. Orbital Tracking: Real-time 3D visualization of satellite trajectories and positions around Earth. Orbital Tracking: Real-time 3D visualization of satellite trajectories and positions around Earth. Orbital Tracking: Real-time 3D visualization of satellite trajectories and positions around Earth.

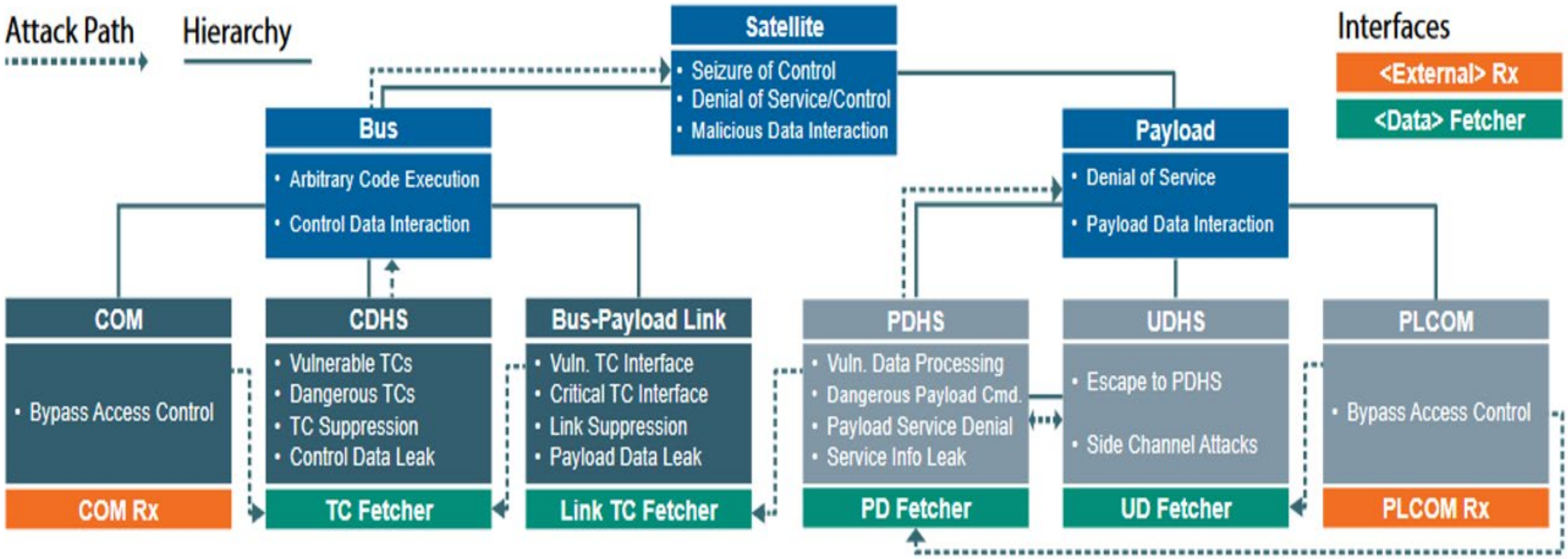
II. Telemetry Monitoring: In-depth tracking of spacecraft health, displaying raw hex streams, live parameter lists, and graphical data trends.

III. Telecommand Management: Tools for scripting, queuing, and executing commands (TCs) sent to the spacecraft.

IV. System Logging: A global event log tracking all system activities, connections, and automated processes.

Satellite Attack Path Hierarchy

Threat taxonomy (or threat model) for satellite firmware, mapping out how cyberattacks can target different internal systems of a satellite.



Satellite Attack Path Hierarchy Explained

Satellite: The Top Level

- Seizure of Control: Taking complete, unauthorized command of the entire satellite and its flight path.
- Denial of Service/Control: Permanently locking out ground operators or completely disabling the satellite so it becomes space junk.
- Malicious Data Interaction: Tampering with the satellite's core functions or faking the primary mission data it sends back to Earth.

<External> Rx

External Rx (Orange Boxes): These represent the physical external receivers (like antennas) on the outside of the satellite. They receive radio frequencies from Earth. An attacker's first step is often beaming a malicious signal here.

The Entry point of an attack

<Data> Fetcher

Data Fetcher (Green Boxes): These are the internal software or hardware buffers that pull the data from the receivers into the satellite's internal systems. They act as the doorways from the outside in.

Bus

- Arbitrary Code Execution: Running unauthorized, malicious software directly on the satellite's core flight computer to hijack its systems.
- Control Data Interaction: Tampering with the critical data that manages the satellite's physical survival, such as its navigation, power, or thermal controls.

Payload

•**Payload:** The mission equipment (like cameras, GPS, or scientific sensors). Attacks here focus on ruining the mission by disabling the instruments or corrupting the data they collect.

COM

COM (Communications): The main radio link to ground control. Attackers target this to bypass security checks and sneak into the system by spoofing a legitimate connection

Bypass Access Control (Between COM Rx & COM): This is the act of sneaking past the satellite's "front door" security. After the antenna (COM Rx) picks up a malicious radio signal from Earth, the attacker must trick, defeat, or bypass the satellite's authentication checks (like encryption or passwords) to successfully break into the main Communications (COM) system

COM rx

External Rx (COM Rx, PLCOM Rx): The physical antennas on the outside of the satellite. This is the very first entry point where a hacker beams a malicious radio signal from Earth.

CDHS

CDHS (Command & Data Handling System): The satellite's "brain." Hackers target it to inject malicious commands, block Earth signals, or steal flight data.

- **Vulnerable TCs**: Sending crafted commands to crash or disrupt the satellite.
- **Dangerous TCs**: Using valid commands to cause harmful actions.
- **TC Suppression**: Blocking ground commands to cut off control.
- **Control Data Leak**: Stealing sensitive onboard data

TC Fetcher

Data Fetcher (TC Fetcher, PD Fetcher, etc.): Internal software or hardware gateways. They act as doorways that pull the malicious data from the outside antennas directly into the satellite's internal computers

Bus-Payload link

Bus-Payload Link: The internal network bridge connecting the vehicle's brain to the mission instruments. Hackers exploit this to "jump" from a compromised Bus over to the Payload.

- Vuln. TC Interface: Exploiting weak internal links to move from the bus to the payload.
- Critical TC Interface: Breaching sensitive pathways to send harmful commands to instruments.
- Link Suppression: Blocking internal communication between bus and payload.
- Payload Data Leak: Intercepting mission data as it moves between systems.

Link TC Fetcher

The internal gateway that forwards commands to instruments; attackers can exploit it to move between systems.

PDHS

PDHS (Payload Data Handling System): Processes the primary mission data (like compressing spy photos). Vulnerabilities here allow attackers to physically damage the sensors or leak sensitive mission intel.

- Vuln. Data Processing: Exploiting bugs in data handling to crash the system.
- Dangerous Payload Cmd.: Using valid commands to physically harm the payload.
- Payload Service Denial: Shutting down or jamming the instruments.
- Service Info Leak: Stealing sensitive mission or instrument data.

PD Fetcher

The internal gateway that pulls commands and data into the main payload system. Hackers exploit this to feed malicious instructions directly to the mission instruments.“

UDHS

UDHS (User Data Handling System): Manages secondary or lower-security user data. Hackers often use this less secure area as entry to break out and invade the main Payload (PDHS).

UD Fetcher

UD Fetcher (User Data Fetcher): The internal gateway that imports secondary user data; due to its lower security level, it is often targeted as an initial entry point into the payload system.

PLCOM

PLCOM (Payload Communications): A dedicated, high-speed radio link used to beam large mission files back to Earth. Hackers try to bypass its access controls to intercept or disrupt the data download

Bypass Access Control

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